

Self-Truncated Velocity Defect

Replacing the Pointwise Lower Gradient Hypothesis

Abstract

The original Plan 2 metric trace used a pointwise lower bound $|\nabla u| \geq m_0$ on every reduced-boundary level surface. That global hypothesis is too strong: tiny components can carry arbitrarily small gradients. This note records the replacement. On levels whose perimeter is already in the Fusco–Julin good class, the low-gradient part of the level surface is discarded relative to the intrinsic average $\bar{f} = \int f/P$, with $f = |\nabla u|$. The discarded part is charged by the Cauchy–Schwarz defect D_H itself. Consequently the velocity defect estimate required by the metric first variation holds without any pointwise lower gradient hypothesis, at the cost of using the isoperimetric-good split that is already present in `WEIGHTEDMETRICTRACE`.

1 Algebraic truncation on one level

Let $(S, \mathcal{H}^{n-1})$ be a finite measure space, let $f : S \rightarrow (0, \infty)$ be measurable, and set

$$P := \mathcal{H}^{n-1}(S), \quad \alpha := \int_S \frac{1}{f} d\mathcal{H}^{n-1}, \quad \beta := \int_S f d\mathcal{H}^{n-1}, \quad D := \alpha\beta - P^2.$$

Write

$$\bar{f} := \frac{\beta}{P}, \quad \phi := \frac{1}{f}, \quad \bar{\phi} := \frac{\alpha}{P}.$$

By Cauchy–Schwarz, $D \geq 0$.

Lemma 1.1 (Low-gradient mass is charged by D). *Let*

$$B := \{x \in S : f(x) < \bar{f}/2\}, \quad G := S \setminus B,$$

and put $\alpha_B := \int_B f^{-1} d\mathcal{H}^{n-1}$. Then

$$\alpha_B \leq \frac{4}{\beta} D, \quad \mathcal{H}^{n-1}(B) \leq \frac{\bar{f}}{2} \alpha_B. \tag{1}$$

Proof. The weighted variance identity is

$$\int_S \frac{(f - \bar{f})^2}{f} d\mathcal{H}^{n-1} = \beta - 2\bar{f}P + \bar{f}^2\alpha = \frac{\beta}{P^2}D. \tag{2}$$

On B one has $\bar{f} - f > \bar{f}/2$, hence

$$\frac{1}{f} \leq \frac{4}{\bar{f}^2} \frac{(f - \bar{f})^2}{f}.$$

Integrating and using $\bar{f} = \beta/P$ in (2) gives $\alpha_B \leq 4D/\beta$. The perimeter estimate follows from $1 \leq (\bar{f}/2)f^{-1}$ on B . $\square$

Lemma 1.2 (Unit-weight reciprocal variance after truncation). *With B, G as above,*

$$\int_S |\phi - \bar{\phi}| d\mathcal{H}^{n-1} \leq \frac{2P}{\beta} D^{1/2} + \frac{20}{\beta} D. \quad (3)$$

Proof. Let $\phi_G := \mathcal{H}^{n-1}(G)^{-1} \int_G \phi d\mathcal{H}^{n-1}$ if $\mathcal{H}^{n-1}(G) > 0$; if $\mathcal{H}^{n-1}(G) = 0$, the estimate below is immediate from Lemma 1.1. On G we have $f \geq \bar{f}/2$, so

$$(\phi(x) - \phi(y))^2 = \frac{(f(x) - f(y))^2}{f(x)^2 f(y)^2} \leq \frac{4P^2}{\beta^2} \frac{(f(x) - f(y))^2}{f(x)f(y)}.$$

Therefore, by the double-integral variance identity on G ,

$$\left(\int_G |\phi - \phi_G| d\mathcal{H}^{n-1} \right)^2 \leq \mathcal{H}^{n-1}(G) \int_G (\phi - \phi_G)^2 d\mathcal{H}^{n-1} \leq \frac{4P^2}{\beta^2} D.$$

It remains to replace the good mean ϕ_G by the global mean $\bar{\phi}$. Since $\phi \leq 2P/\beta$ on G and $\mathcal{H}^{n-1}(B) \leq (\bar{f}/2)\alpha_B$, we have

$$\mathcal{H}^{n-1}(G)|\phi_G - \bar{\phi}| \leq \frac{\mathcal{H}^{n-1}(B)}{P} \int_G \phi d\mathcal{H}^{n-1} + \frac{\mathcal{H}^{n-1}(G)}{P} \int_B \phi d\mathcal{H}^{n-1} \leq 2\alpha_B.$$

Also,

$$\int_B |\phi - \bar{\phi}| d\mathcal{H}^{n-1} \leq \alpha_B + \bar{\phi} \mathcal{H}^{n-1}(B) \leq 3\alpha_B,$$

because

$$\bar{\phi} \mathcal{H}^{n-1}(B) = \frac{\alpha_G + \alpha_B}{P} \mathcal{H}^{n-1}(B) \leq \frac{\mathcal{H}^{n-1}(B)}{P} \frac{2P^2}{\beta} + \alpha_B = \frac{\mathcal{H}^{n-1}(B)}{\bar{f}/2} + \alpha_B \leq 2\alpha_B.$$

Combining these estimates gives

$$\int_S |\phi - \bar{\phi}| d\mathcal{H}^{n-1} \leq \frac{2P}{\beta} D^{1/2} + 5\alpha_B,$$

and Lemma 1.1 gives (3). □

2 Application to torsion level surfaces

Let $\Omega \subset B_R$ have $|\Omega| = \omega_n$, let $u = u_\Omega$, and write $E_\rho = \{u > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$. On a regular level set set

$$S_\rho := \partial^* E_\rho, \quad f := |\nabla u|, \quad P := P(E_\rho), \quad \beta := \int_{S_\rho} f d\mathcal{H}^{n-1} = |E_\rho| = \omega_n \rho^n.$$

Let

$$D_H(t(\rho)) = \left(\int_{S_\rho} \frac{1}{f} d\mathcal{H}^{n-1} \right) \left(\int_{S_\rho} f d\mathcal{H}^{n-1} \right) - P(E_\rho)^2.$$

The normal velocity is

$$V_\rho = \frac{-t'(\rho)}{f}, \quad \bar{V}_\rho = \frac{1}{P(E_\rho)} \int_{S_\rho} V_\rho d\mathcal{H}^{n-1} = \frac{P(B_\rho)}{P(E_\rho)}.$$

The identity

$$(-t'(\rho)) \int_{S_\rho} \frac{1}{f} d\mathcal{H}^{n-1} = P(B_\rho) \quad (4)$$

is the volume-radius derivative of $|E_\rho| = \omega_n \rho^n$.

Proposition 2.1 (Velocity defect without Hyp-G). *Fix $\rho_c \in (0, 1)$ and $P_1 < \infty$. There is $C = C(n, \rho_c, P_1)$ such that, for every regular $\rho \in [\rho_c, 1]$ satisfying $P(E_\rho) \leq P_1$,*

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C(n, \rho_c, P_1) D_H(t(\rho)). \quad (5)$$

No pointwise lower bound on $|\nabla u|$ is assumed.

Proof. Apply Lemma 1.2 on S_ρ with $D = D_H(t(\rho))$. Since $\rho \geq \rho_c$ and $P(E_\rho) \leq P_1$,

$$\beta = \omega_n \rho^n \geq \omega_n \rho_c^n =: \beta_0, \quad P/\beta \leq P_1/\beta_0.$$

Let $\lambda := -t'(\rho) \geq 0$. Then

$$\int_{S_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} = \lambda \int_{S_\rho} |\phi - \bar{\phi}| d\mathcal{H}^{n-1} \leq C_1 \lambda D_H^{1/2} + C_2 \lambda D_H.$$

Talenti's comparison gives $\lambda \leq 1/n$. For the quadratic term, use $D_H \leq \alpha\beta$ and (4):

$$\lambda^2 D_H \leq \lambda^2 \alpha\beta = \lambda P(B_\rho) \beta \leq C(n).$$

Therefore

$$\left(C_1 \lambda D_H^{1/2} + C_2 \lambda D_H \right)^2 \leq C D_H,$$

which proves (5). $\square$

Corollary 2.2 (Fusco–Julin good levels provide P_1). *Let*

$$G_I := \{\rho : D_I(t(\rho)) \leq \eta_I\}.$$

If $\eta_I = \eta_I(n, \rho_c)$ is chosen fixed and small, then $P(E_\rho) \leq P_1(n, \rho_c)$ on $G_I \cap [\rho_c, 1]$. Hence (5) holds on every Fusco–Julin good level in the collar $[\rho_c, \rho_\delta]$.

Proof. Since

$$D_I(t(\rho)) = P(E_\rho)^2 - P(B_\rho)^2,$$

the condition $D_I(t(\rho)) \leq \eta_I$ gives $P(E_\rho)^2 \leq P(B_\rho)^2 + \eta_I \leq n^2 \omega_n^2 + \eta_I$ for $\rho \leq 1$. This gives a uniform P_1 . $\square$

3 Discarded low-gradient sweep

The same algebra makes precise the idea of discarding tiny regions where the gradient is anomalously small.

Corollary 3.1 (The discarded sweep is deficit-controlled). *On a good level $\rho \in G_I \cap [\rho_c, \rho_\delta]$ let*

$$\bar{f}_\rho := \frac{|E_\rho|}{P(E_\rho)}, \quad B_\rho^{\text{low}} := \{x \in \partial^* E_\rho : |\nabla u(x)| < \bar{f}_\rho/2\}.$$

Then

$$\int_{B_\rho^{\text{low}}} V_\rho d\mathcal{H}^{n-1} \leq C(n, \rho_c) (-t'(\rho)) D_H(t(\rho)). \quad (6)$$

Consequently,

$$\int_{G_I \cap [\rho_c, \rho_\delta]} \int_{B_\rho^{\text{low}}} V_\rho d\mathcal{H}^{n-1} d\rho \leq C(n, \rho_c) \delta_T(\Omega). \quad (7)$$

Proof. Equation (6) is Lemma 1.1 multiplied by $\lambda = -t'(\rho)$, using $\beta = |E_\rho| \geq \omega_n \rho_c^n$. Integrating in ρ gives

$$\int (-t'(\rho)) D_H(t(\rho)) d\rho = \int D_H(t(\rho)) d\mu(\rho),$$

which is bounded by $C(n, \rho_c) \delta_T(\Omega)$ by the exact level-set deficit identity on the collar. $\square$

4 Replacement in the metric trace block

In `WEIGHTEDMETRICTRACE`, the metric derivative is estimated on the isoperimetric-good set G_I by

$$|\dot{F}_\rho|_{\mathcal{X}} \leq C_{n,\rho_c} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho,\rho}| d\mathcal{H}^{n-1}.$$

The homothetic trace term is already bounded by the Fusco–Julin radial trace:

$$\left(\int_{\partial^* E_\rho} |H_{z_\rho,\rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C D_I(t(\rho)).$$

Proposition 2.1 replaces the old Hyp-G estimate for the velocity term:

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C D_H(t(\rho)).$$

Therefore, on $G_I \cap [\rho_c, \rho_\delta]$,

$$|\dot{F}_\rho|_{\mathcal{X}}^2 \leq C(n, R, \rho_c) (D_H(t(\rho)) + D_I(t(\rho))).$$

On the complementary isoperimetric-bad set B_I , the existing proof uses the uniform metric Lipschitz bound (A0) and $\mu(B_I) \leq C\delta_T(\Omega)$; this part is unchanged. Thus the integrated action estimate of `WEIGHTEDMETRICTRACE` is recovered without Hyp-G(lower).

Remark 4.1 (Collar choice). The proof only requires a fixed lower radius $\rho_c > 0$ to keep $|E_\rho|$ bounded below and to give an order-one trace window. Since the final boundary-layer transfer uses the endpoint $\rho_\delta = 1 - \kappa\sqrt{\delta_T}$, one may choose, for instance, $\rho_c = 3/4$ once δ_T is small. This makes the repair genuinely collar-local near the outer boundary while preserving an order-one window for the Markov plus kinetic transport argument.

Remark 4.2 (Status). This note does not prove a pointwise lower gradient bound. It shows that such a bound is unnecessary for the Plan 2 metric trace: the portion of a good level where $|\nabla u|$ is too small is self-truncated and paid for by D_H . Tiny remote components are therefore harmless at this step, because their low-gradient sweep contributes to the defect budget rather than to a uniform constant.